

17.3 部分习题参考简答

1. 反之不一定, 例如, $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, |x| < 1$.
2. 收敛半径为1.
3. 收敛域为 $\left[-1 - \frac{1}{k}, -1 + \frac{1}{k}\right]$.
4. $\sum_{n=1}^{\infty} (a_n + b_n)x^n$ 的收敛半径不小于 R .
5. (1) 收敛半径为 $R = +\infty$, 收敛域为 $\mathbb{R}$.
- (2) 收敛半径为 $\sqrt{2}$, 收敛域为 $(-\sqrt{2}, \sqrt{2})$.
- (3) 收敛半径为 $\sqrt[3]{2}$, 收敛域为 $[-\sqrt[3]{2}, \sqrt[3]{2})$.
- (4) 收敛半径为1, 收敛域为 $[-1, 1)$.
- (5) 收敛半径为1, 收敛域为 $[0, 2)$.
- (6) 收敛半径为 $\frac{1}{2}$, 收敛域为 $(-a - \frac{\sqrt{2}}{2}, -a + \frac{\sqrt{2}}{2})$.
- (7) 收敛半径为 $+\infty$, 收敛域为 $\mathbb{R}$.
- (8) 收敛半径为1, 收敛域为 $(-1, 1)$.
6. (1) 收敛域为 $[-1, 1]$ 且 $S(x) = \begin{cases} (1-x)\ln(1-x) + x, & -1 \leq x < 1, \\ 1, & x = 1. \end{cases}$
- (2) 收敛域为 $(-1, 1)$ 且 $S(x) = \frac{3x^3 - x^5}{(1-x^2)^2}, \forall x \in (-1, 1)$.
- (3) 收敛域为 $(-\sqrt{2}, \sqrt{2})$ 且 $S(x) = \frac{2+x^2}{(2-x^2)^2}, \forall x \in (-\sqrt{2}, \sqrt{2})$.
- (4) 收敛域为 $[-1, 1]$ 且 $S(x) = \begin{cases} 0, & x = 0, \\ -1, & x = -1, \\ \frac{(1+x)\ln(1+x)}{x} - 1, & x \in (-1, 0) \cup (0, 1]. \end{cases}$

$$9. \quad (1) \quad \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \frac{1}{4};$$

$$(2) \quad \sum_{n=1}^{\infty} f_n(x) = e^x \ln \frac{1}{1-x}, \quad \forall x \in [-1, 1).$$

11. 收敛区间 $(-2, 4)$.

$$13. \quad f'(x) = \sum_{n=1}^{\infty} \left(\frac{a^n + b^n}{n} \right) x^{n-1}, \quad x \in [-r, r).$$

$$\int_0^x f(t) dt = \sum_{n=1}^{\infty} \left(\frac{a^n + b^n}{n^2(n+1)} \right) x^{n+1}, \quad x \in [-r, r]. \quad \text{其中 } r = \min \left\{ \frac{1}{a}, \frac{1}{b} \right\}.$$